Duty at home

DM9 · Form a commission · 45 min, after S35.

A team consists of five teachers and seven students, all distinct. Four people.

1. How many different commissions can be formed?

2. How many commissions contain exactly two students?

3. How many commissions contain at least one teacher and at least one student?
4. Among the members of a question 2 commission, a chair teacher is appointed and a chair teacher is appointed. student secretary. Count the choices of the commission accompanied by these two powers.
5. By counting the commissions according to their number of teachers, demonstrate that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$